

SOFTWARE REQUIREMENTS SPECIFICATION

for

CareerBridge-AI

AI-Powered Placement & Career Intelligence Ecosystem

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Document Control

Revision History

Version	Date	Description	Author
0.1	TBD	Initial draft derived from project proposal	Project Team
1.0	TBD	First complete SRS release	Project Team

Approval

Role	Name	Signature	Date
Project Guide			
Head of Department			
Project Team Lead			

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for CareerBridge-AI, an AI-powered placement and career intelligence ecosystem. The document is intended for use by the development team, project guides, evaluators, and any stakeholders responsible for designing, implementing, testing, and reviewing the system. It translates the goals described in the original project proposal into verifiable requirements suitable for design and implementation.

1.2 Scope

CareerBridge-AI is a web-based platform that unifies placement-preparation activities that are currently spread across multiple disconnected tools. The system will provide, within a single application:

- AI-driven resume analysis and scoring
- Skill gap identification against target roles
- Personalized learning roadmap generation
- An AI-based virtual career coach
- AI-assisted mock interviews with feedback
- An online coding assessment platform
- A machine-learning based placement probability predictor
- Company-specific preparation content
- GitHub and LinkedIn profile analysis
- A recruiter-facing portal for job posting and candidate discovery
- An alumni mentorship hub
- A 'Digital Career Twin' that models a student's ongoing readiness
- A multi-agent AI backend that coordinates specialized AI agents
- A virtual placement fair for recruiter–student interaction

The system does not cover payroll/HR processing after hiring, learning management for full academic curricula unrelated to placements, or on-campus physical infrastructure (e.g., biometric attendance). These are considered out of scope for this release.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing

Term	Definition
ATS	Applicant Tracking System
API	Application Programming Interface
JWT	JSON Web Token, used for authentication
RBAC	Role-Based Access Control
CGPA	Cumulative Grade Point Average
Digital Career Twin	An AI-maintained profile that mirrors and predicts a student's placement readiness over time
Multi-Agent System	An architecture in which independent, specialized AI agents each handle one functional domain (resume, interview, career, etc.)
Vector Database	A database optimized for storing and querying high-dimensional embeddings (e.g., Pinecone, ChromaDB)

1.4 References

- CareerBridge-AI Project Proposal, Department of Computer Science and Engineering, Academic Year 2026–2027.
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Relevant API documentation for Gemini API, OpenAI API, LangChain, and Sentence Transformers (to be consulted during design).

1.5 Overview

Section 2 describes the product context, users, and constraints. Section 3 lists detailed functional requirements grouped by system module. Section 4 covers external interfaces. Section 5 specifies non-functional requirements. Sections 6 and 7 describe the architecture and data design used to satisfy the requirements. Section 8 lists future enhancements that are explicitly out of scope for the current release.

2. Overall Description

2.1 Product Perspective

CareerBridge-AI is a new, self-contained product. It is not a modification of an existing system, but it is designed to replace the fragmented set of tools (separate resume builders, coding-practice sites, interview-prep apps, and career-guidance portals) that students currently rely on. The product consists of a web frontend, a backend of REST APIs, an AI processing layer that calls large language model providers and hosts ML models, and a persistence layer combining relational and vector databases.

2.2 Product Functions (Summary)

At a high level, the system shall provide the following function groups (each detailed in Section 3):

- Student account management and dashboard
- AI resume analysis and optimization
- Skill gap analysis and learning roadmap generation
- AI career coaching (conversational)
- AI mock interviews with multi-modal feedback
- Coding assessments, contests, and leaderboards
- Placement probability prediction per company
- Company-specific preparation content generation
- GitHub and LinkedIn profile analysis
- Recruiter job posting, candidate search, and analytics
- Alumni mentorship scheduling and tracking
- Digital Career Twin continuous readiness modeling
- Multi-agent orchestration across the above functions
- Virtual placement fair hosting

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Student	Primary user; builds resume, tracks skills, practices interviews, applies to jobs.	Low to Medium
Recruiter	Posts jobs, searches/shortlists candidates, schedules interviews.	Medium
Alumni Mentor	Registers as a mentor, conducts sessions, provides feedback and referrals.	Medium
Placement Cell / Admin	Oversees platform usage, views institution-wide analytics, manages recruiter access.	Medium to High
System Administrator	Manages infrastructure, security configuration, and platform uptime.	High

2.4 Operating Environment

- Client: modern web browsers (Chrome, Edge, Firefox, Safari) on desktop and mobile, via a responsive React.js / Next.js frontend.
- Server: containerized services (Docker) orchestrated with Kubernetes, deployed on AWS.
- AI services: external LLM APIs (Gemini API, OpenAI API) accessed over HTTPS; local ML models served from the backend.
- Databases: PostgreSQL/MySQL for relational data; Pinecone/ChromaDB for vector embeddings.

2.5 Design and Implementation Constraints

- The system must use FastAPI for backend REST services and JWT for authentication, per the approved technology stack.
- AI features are dependent on third-party LLM providers (Gemini API, OpenAI API); availability and rate limits of these providers constrain response time and cost.
- Resume and video data are personally identifiable; all storage and processing must comply with applicable data protection requirements.
- The initial release targets institutional deployment (a single college/university placement cell) rather than a multi-tenant SaaS model.

2.6 Assumptions and Dependencies

- Students have access to a reasonably modern device and stable internet connectivity for video-based mock interviews.
- Recruiters and alumni are willing to create accounts and engage through the platform rather than external channels.
- Third-party AI APIs remain available and within acceptable cost bounds for the institution.
- Historical placement data (for training the placement-prediction model) is available and can be anonymized for use.

3. System Features (Functional Requirements)

This section restates each module from the project proposal as a testable set of functional requirements. Requirement IDs (FR-xxx) are unique across the document and should be used for traceability in test cases and design documents.

3.1 Student Module

Covers registration, authentication, and the student's personal dashboard.

Req. ID	Functional Requirement	Priority
FR-001	The system shall allow a student to register using email or Google authentication.	High
FR-002	The system shall verify a student's email address before granting full access.	High
FR-003	The system shall provide a dashboard summarizing learning progress, resume score, and job-tracking status.	High
FR-004	The system shall allow students to upload and update their resume.	High
FR-005	The system shall generate a downloadable performance report for each student.	High

3.2 AI Resume Analyzer

Parses an uploaded resume and produces a quality/ATS score with suggestions.

Req. ID	Functional Requirement	Priority
FR-006	The system shall parse uploaded resumes in common formats (PDF, DOCX) and extract structured fields (skills, education, experience).	High
FR-007	The system shall compute an ATS compatibility score and an overall resume score out of 100.	High
FR-008	The system shall perform keyword analysis against a target job role.	High
FR-009	The system shall check grammar and formatting and present actionable suggestions (e.g., add measurable achievements, improve summary).	High
FR-010	The system shall recommend relevant certifications or portfolio items based on detected gaps.	High

3.3 Skill Gap Engine

Compares a student's current profile with the requirements of a target role.

Req. ID	Functional Requirement	Priority
FR-011	The system shall accept current skills, certifications, projects, and experience as input.	High
FR-012	The system shall accept a target role and output the list of missing skills relative to that role.	High

Req. ID	Functional Requirement	Priority
FR-013	The system shall rank missing skills by priority and estimated difficulty.	High
FR-014	The system shall estimate the time required to close each skill gap and recommend learning resources.	High

3.4 Personalized Learning Roadmap

Generates a structured, trackable learning plan.

Req. ID	Functional Requirement	Priority
FR-015	The system shall generate daily tasks and weekly milestones based on identified skill gaps.	High
FR-016	The system shall include suggested projects, courses, videos, and assessments in the roadmap.	High
FR-017	The system shall track a student's progress against the roadmap and record achievements.	High
FR-018	The system shall allow the roadmap to be regenerated when the student's target role or skills change.	High

3.5 AI Career Coach

A conversational assistant available at any time.

Req. ID	Functional Requirement	Priority
FR-019	The system shall provide a 24/7 conversational interface for career counseling.	High
FR-020	The system shall track a student's stated goals and provide progress reminders.	High
FR-021	The system shall provide interview guidance and company-specific preparation suggestions on request.	High
FR-022	The system shall log coaching conversations for the student's later reference.	High

3.6 AI Mock Interview

Simulates technical, behavioral, HR, and aptitude interviews with feedback.

Req. ID	Functional Requirement	Priority
FR-023	The system shall support at least four interview types: Technical, Behavioral, HR, and Aptitude.	High
FR-024	The system shall analyze recorded speech for clarity and communication quality.	High
FR-025	The system shall analyze video input to assess non-verbal confidence indicators, where the student has granted camera access.	High

Req. ID	Functional Requirement	Priority
FR-026	The system shall generate a feedback report with improvement suggestions after each session.	High
FR-027	The system shall produce a text transcript of each mock interview session.	High

3.7 Coding Assessment Platform

Provides practice, contests, and company-specific coding questions.

Req. ID	Functional Requirement	Priority
FR-028	The system shall provide an online compiler supporting Python, Java, C++, JavaScript, SQL, and C.	High
FR-029	The system shall provide practice sets, timed contests, and a leaderboard.	High
FR-030	The system shall provide a curated set of company-specific coding questions.	High
FR-031	The system shall automatically evaluate submitted code against test cases and report pass/fail results.	High

3.8 Placement Probability Predictor

Uses an ML model to estimate placement chances per company.

Req. ID	Functional Requirement	Priority
FR-032	The system shall compute a placement probability per company using inputs including CGPA, skills, projects, internships, communication score, coding score, resume quality, and interview scores.	High
FR-033	The system shall display predictions as a percentage per company (e.g., Company X: 82%).	High
FR-034	The system shall update predictions automatically when underlying input data changes.	High
FR-035	The system shall present a disclaimer that predictions are estimates and not guarantees of placement outcomes.	High

3.9 Company Preparation Generator

Produces company-specific preparation material.

Req. ID	Functional Requirement	Priority
FR-036	The system shall generate likely interview questions and a preparation roadmap for a selected company.	High
FR-037	The system shall display eligibility criteria and known hiring patterns for the selected company.	High

Req. ID	Functional Requirement	Priority
FR-038	The system shall surface relevant coding problems and behavioral questions associated with the company.	High
FR-039	The system shall allow students to view anonymized past-experience notes shared by peers or alumni.	High

3.10 GitHub Analyzer

Evaluates a student's public GitHub activity.

Req. ID	Functional Requirement	Priority
FR-040	The system shall analyze a linked GitHub account's repositories, commit frequency, and documentation quality.	High
FR-041	The system shall compute a GitHub score and a developer profile rating.	High
FR-042	The system shall recommend improvements (e.g., adding README files, increasing contribution frequency).	High

3.11 LinkedIn Optimizer

Analyzes a student's LinkedIn profile for completeness and effectiveness.

Req. ID	Functional Requirement	Priority
FR-043	The system shall analyze headline, summary, skills, experience, and certifications on a linked LinkedIn profile.	High
FR-044	The system shall compute a LinkedIn score and provide optimization tips.	High
FR-045	The system shall surface relevant industry insights based on the student's target role.	High

3.12 Recruiter Portal

Enables recruiters to post jobs and evaluate candidates.

Req. ID	Functional Requirement	Priority
FR-046	The system shall allow recruiters to post job openings with role, required skills, and eligibility criteria.	High
FR-047	The system shall allow recruiters to search and filter candidates using AI-based matching.	High
FR-048	The system shall allow recruiters to shortlist candidates and schedule interviews through the platform.	High
FR-049	The system shall provide recruiters with an analytics dashboard summarizing applicant pipeline and quality.	High

3.13 Alumni Mentorship Hub

Connects students with alumni mentors.

Req. ID	Functional Requirement	Priority
FR-050	The system shall allow alumni to register as mentors and specify availability.	High
FR-051	The system shall allow students to book mentorship sessions and receive resume reviews.	High
FR-052	The system shall allow mentors to record session feedback and status.	High
FR-053	The system shall support mentors providing referrals to students, where the mentor opts to do so.	High

3.14 Digital Career Twin

A continuously updated AI model of each student's readiness.

Req. ID	Functional Requirement	Priority
FR-054	The system shall maintain a continuously updated profile tracking a student's skills, performance, growth, and goals.	High
FR-055	The system shall predict future placement readiness based on trend data.	High
FR-056	The system shall proactively surface recommendations as the student's profile changes.	High

3.15 Multi-Agent AI System

Coordinates specialized AI agents behind the scenes.

Req. ID	Functional Requirement	Priority
FR-057	The system shall implement independent agents for resume, interview, career, learning, placement, mentorship, networking, and analytics functions.	High
FR-058	The system shall route each user request to the appropriate agent(s) and combine outputs where more than one agent is involved.	High
FR-059	The system shall log agent decisions sufficiently to support debugging and auditability.	High

3.16 Virtual Placement Fair

Hosts an online event connecting recruiters and students.

Req. ID	Functional Requirement	Priority
FR-060	The system shall allow companies to host virtual booths during a scheduled placement fair.	High

Req. ID	Functional Requirement	Priority
FR-061	The system shall support live recruiter sessions and real-time interviews during the fair.	High
FR-062	The system shall support online assessments and AI-based candidate-company matching during the event.	High
FR-063	The system shall provide a digital networking feature allowing students and recruiters to connect during the fair.	High

4. External Interface Requirements

4.1 User Interfaces

- A responsive web application built with React.js, Next.js, Tailwind CSS, and Framer Motion, supporting desktop and mobile browsers.
- Role-specific dashboards for Students, Recruiters, Mentors, and Administrators.
- Accessible design following common WCAG-style guidelines (readable contrast, keyboard navigation) as a target, not a hard constraint for the first release.

4.2 Hardware Interfaces

- Camera and microphone access on the student's device for AI mock interview video/speech analysis (optional; text-only mode shall remain available).
- No specialized hardware is required for other modules.

4.3 Software Interfaces

Interface	Purpose
Gemini API / OpenAI API	Natural-language generation, resume analysis, career coaching, question generation
LangChain / LlamaIndex	Orchestration of LLM calls, retrieval-augmented generation for company/role-specific content
Sentence Transformers	Generating embeddings for skills, resumes, and job descriptions
Pinecone / ChromaDB	Vector storage and similarity search over embeddings
PostgreSQL / MySQL	Relational storage of students, jobs, interviews, and mentorship records
GitHub API	Reading public repository and commit data for the GitHub Analyzer
LinkedIn (via user-provided data or supported API)	Reading profile fields for the LinkedIn Optimizer

4.4 Communication Interfaces

- All client-server communication shall use HTTPS.
- The backend shall expose REST APIs secured with JWT-based authentication.
- Real-time features (e.g., the virtual placement fair, live chat with the AI coach) shall use WebSocket or an equivalent low-latency protocol.

5. Non-Functional Requirements

5.1 Performance Requirements

- Standard page loads (dashboard, resume view) shall complete within 3 seconds under normal load.
- AI-generated responses (resume feedback, coach replies) shall be returned within 10 seconds for typical inputs; long-running analyses (e.g., full mock interview report) may take longer but shall show progress indicators.
- The coding assessment compiler shall return execution results within 5 seconds for typical submissions.

5.2 Security Requirements

- The system shall use JWT authentication for all API access.
- Passwords shall be stored using a strong, salted hashing algorithm; plaintext passwords shall never be stored.
- The system shall implement role-based access control (RBAC) so that students, recruiters, mentors, and administrators only access data appropriate to their role.
- Uploaded files (resumes, videos) shall be validated and scanned before storage.
- Sensitive data (resumes, video recordings, personal identifiers) shall be encrypted at rest and in transit.
- The system shall maintain session management with token expiry and secure logout.
- All user inputs shall be validated and sanitized to prevent injection attacks.

5.3 Reliability & Availability

- The system shall target 99% uptime during active placement seasons.
- The system shall gracefully degrade AI-dependent features (e.g., show a cached or simplified response) if a third-party AI provider is unavailable.

5.4 Usability

- New students shall be able to complete registration and generate an initial resume score within 10 minutes without external help.
- The interface shall provide clear, jargon-free explanations alongside AI-generated scores and predictions.

5.5 Scalability & Maintainability

- The system shall be deployed using Docker containers orchestrated by Kubernetes to allow horizontal scaling during peak placement periods.
- AI agents (Section 3.15) shall be independently deployable and updatable without requiring a full-system redeployment.
- The system shall log key events (logins, AI calls, errors) to support monitoring and troubleshooting.

5.6 Portability

- The web client shall function correctly on the latest two major versions of Chrome, Firefox, Edge, and Safari.

6. System Architecture

CareerBridge-AI follows a layered architecture. Each layer communicates with the layer below it through well-defined interfaces, allowing components (e.g., the AI provider or the vector database) to be replaced with minimal impact on other layers.

Layer	Technologies	Responsibility
Frontend	React.js, Next.js, Tailwind CSS, Framer Motion	Renders role-specific UI, handles client-side interaction and state
Backend	FastAPI, REST APIs, JWT	Exposes business-logic APIs, enforces authentication and authorization
AI Layer	Gemini API, OpenAI API, Sentence Transformers, LangChain, LlamaIndex, ML models	Performs resume analysis, coaching, interview evaluation, and prediction tasks
Vector Database	Pinecone, ChromaDB	Stores embeddings for semantic search (e.g., matching resumes to job descriptions)
Database Layer	PostgreSQL, MySQL	Stores structured, transactional data (students, jobs, interviews, mentorship)
Cloud Infrastructure	AWS, Docker, Kubernetes, S3	Hosts, scales, and stores files for the platform

Data flows from the Frontend to the Backend over authenticated REST calls. The Backend delegates AI-specific tasks to the AI Layer, which may call external LLM providers or local models, and reads/writes embeddings to the Vector Database and structured records to the Database Layer. The Multi-Agent AI System (Section 3.15) sits within the AI Layer and coordinates specialized agents for each functional domain.

7. Data Requirements

The following core entities shall be persisted by the system. Field lists are illustrative and will be refined during detailed design.

Students

Field	Notes
student_id (PK)	
name	
email	
department	
cgpa	
skills	
resume_score	
placement_score	

Resumes

Field	Notes
resume_id (PK)	
student_id (FK)	
file_path	
ATS_score	
upload_date	

Skills

Field	Notes
skill_id (PK)	
student_id (FK)	
skill_name	
proficiency	

Interviews

Field	Notes
interview_id (PK)	
student_id (FK)	
feedback	
score	
communication_score	

Jobs

Field	Notes
job_id (PK)	
company	
role	
required_skills	
eligibility	

Mentorship

Field	Notes
mentor_id (PK)	
student_id (FK)	
session_date	
feedback	
status	

8. Other Requirements & Future Enhancements

The following items are recognized as valuable but are explicitly out of scope for the initial release. They are listed here for planning purposes and should not be treated as current requirements.

- AI-based video interview analysis beyond the baseline confidence/communication assessment
- Real-time, deep recruiter system integration (e.g., direct ATS syncing)
- An internship marketplace
- A global (cross-institution) mentorship platform
- Blockchain-based certificate issuance and verification
- A standalone AI hiring assistant for recruiters
- A university-wide analytics dashboard spanning multiple institutions
- AI-based employment demand forecasting
- Enterprise partnership integrations

Social Impact (Context)

The proposal frames CareerBridge-AI as a way to democratize access to career guidance, offering more affordable mentoring and more consistent placement support regardless of a student's background. While not a testable requirement, this framing is retained here to guide prioritization decisions (e.g., favoring features that widen access over features that primarily add convenience).

Appendix A: Glossary

See Section 1.3 for the full list of terms, acronyms, and abbreviations used in this document.

Appendix B: Notes on Traceability

Each functional requirement in Section 3 carries a unique identifier (FR-001, FR-002, ...) intended for use in the test plan and traceability matrix. Non-functional requirements in Section 5 should be referenced by their subsection number (e.g., NFR-5.2 for security) until a similar ID scheme is formally adopted.